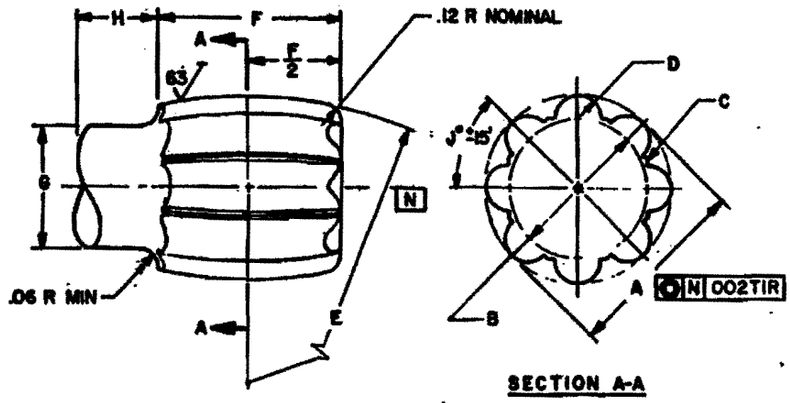
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METRIC CONVERSION

1.00 in. = 25.4 mm



SECTION A-A

CIRCULAR SPLINE

SPLINE DESIGN DATA

BASIC SIZE	NO. OF TEETH	SPLINE DIA. A	PORM DIA. B	ROOT DIA MAX. C	TOOTH RADIUS D	CROWN RADIUS E	SPLINE LENGTH F	SHAFT DIA. MAX. G	SHAFT LENGTH MIN. H	TOOTH INDEX ANGLE J
3/4	6	0.600 0.598	0.413 BASIC	0.470	0.093 0.091	10.0 8.0	0.85 0.65	0.438	0.31	60°
1	8	1.000 0.998	0.750 BASIC	0.800	0.124 0.122	15.0 13.0	1.10 0.90	0.670	0.44	45°
1 1/2	8	1.250 1.248	0.937 BASIC	1.000	0.155 0.153	16.0 14.0	1.35 1.15	0.910	0.44	45°

NAVY DESIGN STANDARD FOR CIRCULAR SPLINE COUPLINGS USED IN AIRCRAFT ELECTROMECHANICAL ACCESSORIES.

© REVISED AND REDRAWN

APPROVED 6 MAY 1978 REVISED (A) 31 JULY 1977 (C) 13 JUNE 1980

P.A. NAVY – AS Other Cust	TITLE CIRCULAR SPLINE & ADAPTER DETAILS ENGINE DRIVEN ACCESSORIES	MILITARY STANDARD MS14169 (AS)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 3

DD FORM 1 MAR 72 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. Project No. 5115-N449 PLATE NO. 23071

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SPLINE ADAPTER DESIGN DATA

BASIC SIZE	INTENDED MATING SPLINE	NO. OF TEETH	MIN. LENGTH A	MAJOR DIA. B	ROOT DIA. C	SPLINE DIA. D	MINOR DIA. E	FORM DIA. F	TOOTH RADIUS G	COUNTER BORE DIA. H	INTER. INDEX ANGLE J	EXTER. INDEX ANGLE K	NOM. ROOT WIDTH L	OVER WIRES DIA. M	WIRE DIA. N
3/4	0.800P.D. (20/30D.P.)	6 int. 16 ext.	1.50	0.817 0.812	0.733 0.725	0.604 0.602	0.475 0.473	0.413 BASIC	0.095 0.094	0.63 0.61	60°	22° 30'	0.038	0.936 0.933	0.0900
3/4	0.800P.D. (20/40D.P.)	6 int. 16 ext.	1.50	0.817 0.812	0.750 0.742	0.604 0.602	0.475 0.473	0.413 BASIC	0.095 0.094	0.63 0.61	60°	22° 30'	0.044	0.935 0.932	0.0900
1	1.200P.D.	8 int. 24 ext.	1.75	1.270 1.265	1.124 1.113	1.004 1.002	0.811 0.809	0.750 BASIC	0.126 0.125	1.03 1.01	45°	15° 0'	0.033	1.418 1.415	0.12000
1-1/2	1.625P.D.	8 int. 26 ext.	2.00	1.700 1.695	1.554 1.543	1.256 1.252	1.015 1.013	0.937 BASIC	0.157 0.156	1.28 1.26	45°	13° 51'	0.056	1.842 1.838	0.13091

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P.A. NAVY – AS Other Cust	TITLE CIRCULAR SPLINE & ADAPTER DETAILS ENGINE DRIVEN ACCESSORIES	MILITARY STANDARD MS 14169 (AS)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF 3

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NOTES

1. DIMENSIONS
ALL LINEAR DIMENSIONS ARE IN INCHES EXCEPT ROUGHNESSES WHICH ARE IN MICROINCHES.

(C) 2. APPLICATION
THE SPECIFIED ADAPTERS MAY BE USED ONLY WITH INVOLUTE SPLINES HAVING 30 DEGREES PRESSURE ANGLES.

3. SPLINES
MALE SHAFT SPLINES SHALL HAVE A MINIMUM SURFACE HARDNESS OF 34 ROCKWELL C TO A MINIMUM ANGLES.

4. FINISH
ALL BURRS AND SHARP EDGES SHALL BE REMOVED.

5. APPLICATION
CIRCULAR SPLINE COUPLINGS MAY BE USED AS REPLACEMENTS FOR WORN INVOLUTE SPLINE COUPLINGS WITHOUT MODIFICATION OR REPLACEMENT OF THE AIRCRAFT ACCESSORY DRIVE PAD SHAFT (FEMALE SPLINE). PRIOR TO RETROFIT, THE FEMALE SPLINE SHALL BE THOROUGHLY CLEANED TO REMOVE SEDIMENT AND GREASE. THE DESIGN RESULTS IN AN INTERFERENCE FIT OF THE SPLINE ADAPTER INTO THE INVOLUTE SPLINE AND A SLIDING FIT OF THE CIRCULAR SPLINE SHAFT INTO THE ADAPTER.

(C) 6. INSTALLATION
THE NONMETALLIC SPLINE ADAPTER IS DESIGNED TO BE INSTALLED IN THE DRIVE SHAFT FEMALE SPLINE PRIOR TO INSTALLING THE DRIVEN ACCESSORY. NONMETALLIC SPLINE ADAPTERS ARE INTENDED TO FIT TIGHTLY IN THE FEMALE SPLINE IN WHICH THEY ARE INSTALLED. TO AVOID DAMAGING THE DRIVE SHAFT OR ACCESSORY SHAFT BEARINGS AND SEALS, THE SPLINE ADAPTERS SHOULD BE PRESSED INTO THE MATING SPLINES AND THE PRESSURE SHOULD BE REMOVED AS SOON AS THE SPLINE ADAPTER IS FULLY INSTALLED.

7. RETENTION AND PRELOAD
THE NONMETALLIC SPLINE ADAPTERS HAVE BEEN DESIGNED TO FIT TIGHTLY IN NEW FEMALE SPLINES. THE TIGHTNESS OF THE FIT WILL DIMINISH WHEN THE NONMETALLIC ADAPTER IS INSTALLED IN WORN OR USED FEMALE SPLINES. THE SPLINE ADAPTER HAS BEEN DESIGNED TO FIT TIGHTLY TO ACHIEVE A COMPRESSIVE PRELOAD WHICH EXTENDS THE OPERATING TORSIONAL RANGE OF THE NONMETALLIC SPLINE ADAPTER. THE TIGHT FIT ALSO PROVIDES A RESISTANCE TO AXIAL MOVEMENT WHICH AFFORDS SOME DEGREE OF RETENTION. HOWEVER, THE APPLICATION GENERALLY SHOULD REQUIRE SOME POSITIVE METHOD OF AXIAL RETENTION OF THE PART. POSITIVE RETENTION CAN BE ACHIEVED BY A VARIETY OF METHODS INCLUDING LENGTHENING THE NONMETALLIC ADAPTER, INCORPORATION OF FLANGES IN THE ADAPTER DESIGN, AND THE USE OF SNAP OR SPIRAL LOCK RINGS TO LIMIT THE AMOUNT OF AXIAL CLEARANCE.

8. MATERIAL SELECTION
THE PLASTIC SPLINE ADAPTER SHALL BE FABRICATED FROM HIGH STRENGTH, SELF-LUBRICATING POLYMERIC MATERIALS HAVING ULTIMATE COMPRESSIVE STRENGTHS OF A MINIMUM OF 28,000 PSI. TYPICAL MATERIALS INCLUDE THE POLYIMIDE, ARAMID AND POLYAMIDE-IMIDE RESINS, BELONGING TO THE DUPONT VESPEL AND AMOCO TORLON FAMILIES. THE PROCESS OF MATERIAL SELECTION SHOULD CONSIDER STRENGTH, FATIGUE AND AGING PROPERTIES, FLUID COMPATIBILITY, THERMAL EXPOSURE, DIMENSIONAL STABILITY, AND FORMING PROPERTIES.

9. SHEAR SECTION
THE SHAFT SHALL INCLUDE A SHEAR SECTION ALLOWING THE SHAFT TO SHEAR AT THE TORQUE SPECIFIED BY THE SPECIFICATION FOR THE ACCESSORY EQUIPMENT.

10. PATENT RESTRICTION
UNDER THE CONDITIONS OF PATENT NUMBER 3,620,043, "SPLINE-TYPE PIVOTS, UNIVERSAL JOINTS AND FLEXIBLE COUPLINGS", DATED NOVEMBER 16, 1971, ARINC RESEARCH CORPORATION HAS GRANTED TO THE UNITED STATES GOVERNMENT A ROYALTY-FREE LICENSE TO USE THE DESCRIBED CIRCULAR SPLINE COUPLING DESIGN FOR GOVERNMENTAL PURPOSES ONLY.

☆U.S. GOVERNMENT PRINTING OFFICE: 1980 — 603-121/2910

APPROVED 6 MAY 1976 REVISED (C) FOR CHANGES SEE SHEETS 1 THRU 3

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P.A. NAVY – AS Other Cust	TITLE CIRCULAR SPLINE & ADAPTER DETAILS ENGINE DRIVEN ACCESSORIES	MILITARY STANDARD MS 14169 (AS)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 3 OF 3

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